

IntelliCID: Intelligent Caustics Illumination Device

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ABSTRACT

Environmental illumination has the power to influence our daily lives to some extent. When combined with the inherent human affinity for nature, the presence of natural illumination, such as caustics and *komorebi*, can be used to create a relaxing environment. However, if this illumination does not seamlessly adapt to our daily activities, it may become overwhelming and unpleasant. To address this issue, we present IntelliCID, an intelligent and interactive system designed for smart environments. This system is capable of sensing the user's state and adjusting the intensity of the dynamic caustics lighting effect accordingly.

CCS CONCEPTS

• **Human-centered computing** → **Displays and imagers; Interaction devices; Human computer interaction (HCI); Displays and imagers.**

KEYWORDS

Caustics, smart environment, interactive device, illumination

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1 INTRODUCTION

We are surrounded by various sources of illumination. The Sun provides light outdoors, while streetlights, room lights, and projectors make indoor environments visible to the human eye. Extensive research has validated the effects of indoor lighting on cognitive performance [5] and psychological well-being [6]. As a result, lighting design has become a dedicated area in the field of engineering and product design. This impact is often combined with the human tendency to feel at ease when connected with nature [9]. For example, ambient, natural lighting, such as the dappled light that filters through gaps between leaves (known as *komorebi* in Japanese), can bring relief to our minds. Another type of ambient lighting is caustics, which are patterns created when light rays are reflected or refracted by curved surfaces or objects. These patterns result in areas of concentrated light and brightness, often producing intricate

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and visually appealing effects. Caustics are commonly observed in environments where water is present, such as the ripples and waves of a swimming pool, the surface of a bathtub, or the sea.

Caustics are abstract and peculiar patterns that ambiently represent the presence of water. We believe that dynamic caustics lighting is one type of light that enables us to feel closer to nature and has a positive psychological effect on relaxation.

However, the beauty and attractiveness of the dynamic caustics are apt to stand out too much when you do everyday activities indoors. For example, when you read books, the effect distracts you if the pattern frequently changes. Plus, if you want to control the effect dynamically and finely, the device cost will get higher. To tackle this problem of installing dynamic caustics illumination in daily life, we propose IntelliCID, the illumination device that is **(1)cost-efficient to be set up in real environments, (2)able to control the intensity of the effect according to the user's state, such as resting or active.** We realize this device by combining a microcontroller, human body sensing module, and vibrators. By installing this device, you can feel nature in your daily room.

2 RELATED WORKS

Caustics have received attention for a long time. In the field of computer graphics, caustics are one of the targets to be reproduced by realistic rendering[2, 3] and to control the pattern[8]. However, CG reproduction is inferior to the real ones when projected onto a real surface. In terms of physically controlling the caustics, recent work [7] can dynamically control patterns that are unachievable with static refractive surfaces using a method that combines acoustofluidics with optics. This setup which uses such as the Phased Array Transducers, is expensive and requires a large space to be prepared in everyday indoor room environments.

Besides being a work in the field of research, Leandro Erlich's swimming pool [1] installed at the 21st Century Museum of Contemporary Art in Kanazawa, Japan, attracts so many customers because of its photogenic caustics effect. Just like creating other works, several methods have been proposed to structurally create the caustics effect in the field of architecture[4]. However, these large setups are not easily implemented in typical homes and cannot adjust the effect according to the user's preference.

None of the methods and works mentioned above aim to bring the dynamic caustics effect indoors in a cost-friendly way while also allowing for the modulation of intensity in a smart environment.

3 PROPOSED SYSTEM: INTELLICID

We propose a cost-effective, dynamically interactive system for generating caustic effects indoors. This system utilizes microcontrollers, sensors, vibrators, and speakers. The system detects the user's posture and activates actuators to create liquid waves. By passing light through these liquid waves, dynamic caustic patterns

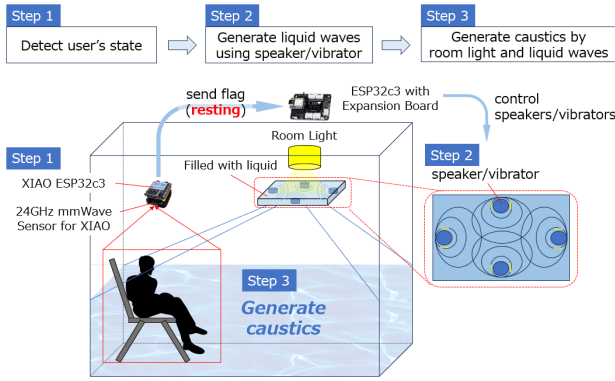


Figure 1: Proposed system: resting state

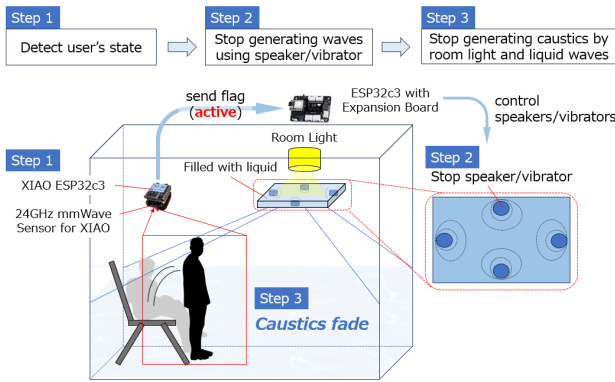


Figure 2: Proposed system: active state

are projected. Providing an interactive caustic effect is expected to positively impact the user. For instance, simulating an underwater caustic effect for a relaxed user is expected to enhance relaxation and reduce stress.

3.1 System Overview

The proposed system is designed for the dynamic caustics effect in indoor rooms. The workflow of the system is as follows:

(1) User's state detection

The sensor continuously monitors the user's physiological signals to detect their state. Upon detecting the user's state, a signal flag is sent to a microcontroller.

(2) Signal Transmission and Actuation

The microcontroller receives the relaxation signal and activates actuators in liquid, such as vibrators or speakers. These actuators are used to generate waves on the surface of the liquid.

(3) Caustics Projection

A room light is positioned to shine on the liquid surface, where the waves have been generated. The interaction of light with the liquid waves creates caustics patterns, which are projected onto the room's walls and ceiling. This projection creates an immersive visual effect.

Figures 1 and 2 illustrate the system architecture. Figure 1 represents the system when the user is resting, while Figure 2 represents the system when the user is in an active state. In the resting state, caustics are generated by creating waves using an actuator. By simulating caustics indoors, the system successfully evokes the sensation of being underwater, thereby enhancing relaxation and reducing stress. In the active state, unlike the resting state, the actuator is stopped, and the system is controlled to calm the waves. Users can control the caustics with simple actions, either by resting or being active.

3.2 System Component

Below is a list of the components of the proposed system.

- Seed Studio XIAO ESP32C3 × 2: Used to send relaxing signal to another one and control actuators.
- 24GHz mmWave Sensor for XIAO × 1: Used to detect whether the user is in a relaxed posture.
- Seed Studio Expansion Board Base for XIAO with Grove OLED × 1: Used to control actuators with ESP32c3.
- Grove - Speaker × 4: Used to generate waves in liquid. They're inserted into the liquid through a thin rubber cover.
- Grove - Vibration Motor × 4: Used to generate waves in liquid. They're inserted into the liquid through a thin rubber cover.

4 CONCLUSION

The IntelliCID system introduces a novel approach to indoor environmental design by leveraging dynamic caustics illumination. This system is designed to be cost-efficient and easy to integrate, using microcontrollers, sensors, and actuators to generate and control caustic lighting effects based on the user's state, whether they are resting or active. This dynamic control not only promotes relaxation and stress reduction but also avoids potential distractions caused by constant visual effects during everyday activities.

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